Split-prime pattern and the ECDLP: a scalar-only obstruction, an independent recount, and the routes that stay open

research/split_prime_ecdlp_20261107 · 2026-11-07 · branch `claude/novel-ecdlp-algorithm-iams4j`

Status: derived obstruction plus a native recount; no algorithm claim. **Result class:** boundary study and lead closure, not a cost reduction for any ECDLP method. `S` and end-to-end cost are unset; no leaderboard, benchmark or rho comparison is touched. Claim record for the recount: `PCC-20261107-3403f9` (kind statement; standing unverified until CI signs it) [pcc:unverified]. Problem record: `problems/split-prime-pattern-scalar-only.md`.

1. Question and answer

Request (user, 2026-11-07): a novel ECDLP algorithm based on something unique and not tried before, for example the prime-splitting pattern observed in the published prime curves (uploaded report `split_prime_report.pdf`, sha256 `d38be158…7a0c8`, "Split-prime deficit in published NIST / X9.62 prime curves").

Answer. The pattern is real: recounted here from the curve parameters, every number agrees with the report (§3). It cannot feed an ECDLP algorithm below the generic bound on these curves: for every prime $\ell \neq n$, the objects it governs (ℓ-torsion, Frobenius eigenlines, the two rational ℓ-isogenies of a split ℓ, the class group of the Frobenius order, the endomorphism ring) reach the prime-order group $E(\mathbb{F}_p)$ only as multiplication by a known scalar or as an isomorphism that preserves the discrete logarithm (§2). This agrees with the lab's isogeny-class invariance framing (`research/isogeny_class_difficulty_20261008`) and with the measured null on index-calculus yield across a class (`research/isogeny_class_ic_yield_20261102`). The pattern's one ECDLP-adjacent consequence is a lower degree of the horizontal expander used by random self-reduction (§4). Candidate directions that survive the obstruction, with their cheapest tests, are in §5; none is a claim.

Scope. The ten prime-order curves prime192v1 (P-192), prime192v2, prime192v3, secp224r1 (P-224), prime239v1/v2/v3, prime256v1 (P-256), secp384r1 (P-384), secp521r1 (P-521); p and n from `openssl ecparam -param_enc explicit` (OpenSSL 3.0.13), cofactor 1. Objects: $t = p + 1 - n$, $\Delta = t^2 - 4p$, $K = \mathbb{Q}(\sqrt{\Delta})$, and for each odd prime ℓ the Kronecker symbol $\chi(\ell) = \left(\frac{\Delta}{\ell}\right)$: split (+1, Elkies), ramified (0), inert (−1, Atkin).

Sources checked. KB unavailable (the `research-kb` MCP server is not loaded; `scripts/research-kb` finds no executable; the lead atlas checkout is absent). Repository paths read: `research/isogeny_class_difficulty_20261008/README.md`, `research/isogeny_class_ic_yield_20261102/p2_prime_analogues/RESULTS.md` and `report/REPORT.md`, `problems/prime-scalar-exact-splitting.md`, `problems/tree-glued-pdp-separator-coupling.md`, `problems/curve25519-family-algebraic-ic-tower.md`, `docs/endomorphism-rules.md`. Literature from memory, primary sources not re-read this session: Lenstra 1996 ($E(\mathbb{F}_q) \cong \mathcal{O}/(\pi - 1)$ as $\mathcal{O}$-modules), Jao–Miller–Venkatesan 2005 (random self-reducibility across an isogeny class via expander walks over split ℓ, under GRH), Shoup 1997 ($\Omega(\sqrt{n})$ generic bound), Schoof–Elkies–Atkin (split ℓ speed up point counting only).

2. The obstruction (derived; no Lean proof yet)

Setting. $E/\mathbb{F}_p$ ordinary, $\#E(\mathbb{F}_p) = n$ prime, cofactor 1, Frobenius π with $\pi^2 - t\pi + p = 0$, $\mathcal{O} = \mathrm{End}(E) \supseteq \mathbb{Z}[\pi]$, $N = (\pi - 1)$ the ideal of $\mathcal{O}$ of norm n.

Theorem (scalar-only action). Let ℓ be a prime different from n and p. Every $\mathbb{F}_p$-rational isogeny $\varphi : E \to E'$ of ℓ-power degree (each horizontal ℓ-isogeny of a split ℓ, their compositions around the class group, every vertical ℓ-isogeny) and every endomorphism $\alpha \in \mathcal{O}$ satisfy:

1. φ restricted to $E(\mathbb{F}_p)$ is a group isomorphism onto $E'(\mathbb{F}_p)$ and $\varphi([k]P) = [k]\varphi(P)$: a transported DLP instance has the same answer.
2. α acts on $E(\mathbb{F}_p)$ as multiplication by $s_\alpha = \alpha \bmod N \in \mathbb{Z}/n$, computable from the coordinates of α in $\mathcal{O}$ (for $\alpha = u + v\pi$, $s_\alpha = u + v$).
3. The Galois module $E[\ell]$, its Frobenius eigenvalues λ, μ, the fiber polynomials of φ, and every ℓ-pairing value involving a point of $E(\mathbb{F}_p)$ are independent of which point of $E(\mathbb{F}_p)$ is considered.

Proof sketch. (1) $\deg\varphi$ is prime to n, so $\ker\varphi \cap E(\mathbb{F}_p) = \{O\}$ and φ is injective on the finite group $E(\mathbb{F}_p)$; $E'(\mathbb{F}_p)$ has the same order, so φ is onto; it is a homomorphism, so it commutes with $[k]$. (2) $E(\mathbb{F}_p)$ is cyclic of prime order, so

every endomorphism of it is a scalar; which scalar follows from Lenstra's isomorphism $E(\mathbb{F}_p) \cong \mathcal{O}/N$ under which $\pi \mapsto 1$. (3) $E[\ell]$ and $E(\mathbb{F}_p)$ are Galois modules of coprime order; a bilinear pairing $t(T, P)$ with ord $T = \ell$, ord $P = n$ satisfies $t^\ell = t^n = 1$, so $t = 1$; the fiber of φ above $\varphi(P)$ is $P + \ker \varphi$ with the Galois structure of $\ker \varphi$. ■

Corollary. An algorithm whose only operations are the group law, the maps above and point-independent Galois data is a generic-group algorithm with scalar oracles; Shoup's bound gives $\Omega(\sqrt{n})$ group operations, and each map costs at least one group operation to evaluate on a point. So the split-prime pattern alone, for any cutoff and any deficit, cannot yield a sub-generic algorithm on a prime-order curve; by the JMV self-reduction every curve reachable along the horizontal edges that the split primes provide is exactly as hard.

Not covered. Anything that reads $\mathbb{F}_p$-coordinates non-linearly: summation polynomials, division polynomials, and (for small embedding degree only, which these curves lack) the n-pairing. These are the lab's index-calculus tracks. The pattern can enter them indirectly by choosing which curve of the class to decompose on; the lab measured that choice null on 2026-11-02 (P2: ICC not distinguishable from zero at 24–32 bits, planted-20 % power 0.84–1.00 in every informative cell, one suspected curve effect not replicated).

The reformulation that looks like a way in. By Shimura reciprocity the DLP k is the element of $\mathrm{Gal}(K_N/H) \cong (\mathcal{O}/N)^\times/\pm 1$ mapping the canonical-lift torsion point $\tilde{P}$ to $\tilde{Q}$, and the Artin symbol of a principal ideal (α) acts as $\alpha \bmod N$. Writing k as a product of known scalars s_α of smooth principal α is a multiplicative DLP in $\mathbb{F}_n^\times$, feasible by index calculus — but only on the residue k, which is the unknown. The group is presented by coordinates, never by residues in $\mathcal{O}/N$, so "index calculus in $\mathcal{O}$" has no first step; enumerating smooth α and comparing $\alpha(P)$ with Q is baby-step giant-step with an unusual giant-step set and the same $\sqrt{n}$ cost.

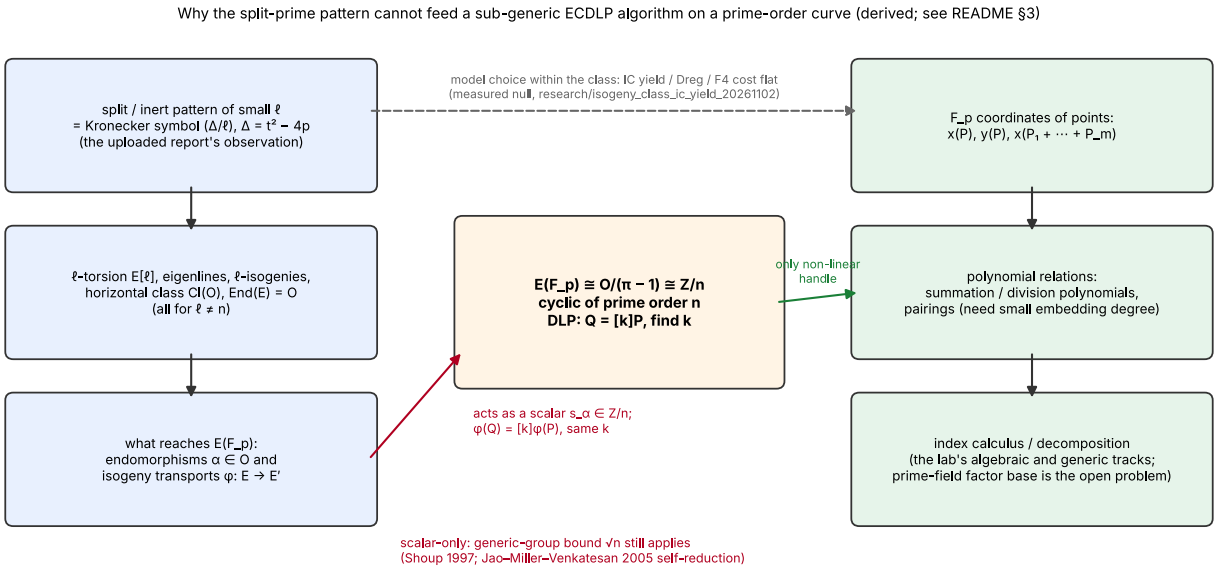


Figure 1: Where the pattern lives (left), the group it must reach (centre), and the only non-linear handle on that group (right). Derived; the dashed link is the route measured null on 2026-11-02.

3. Independent recount of the deficit (measured, native)

Method. `tools/splitcount.rs` (dependency-free Rust, `rustc -O`, 0.3 s). For each odd prime $\ell \le 200000$ it reduces the hex parameters p, n modulo ℓ, forms $t \equiv p + 1 - n$, $\Delta \equiv t^2 - 4p$, and takes the Legendre symbol by Euler's criterion. Baseline (the report's model, re-derived in the source comment): Frobenius mod ℓ uniform among matrices of $\mathrm{GL}_2(\mathbb{F}_\ell)$ with determinant p, so $P(\text{split}) = (\ell - 2 - \chi(p))/(2(\ell - 1))$, $P(\text{ramified}) = (1 + \chi(p))\ell/(\ell^2 - 1)$; the trace class $\tau \equiv p + 1$ (which would give $\ell \mid n$) is removed and the rest renormalised. $z = (\text{observed} - \sum P)/\sqrt{\sum P(1 - P)}$ over odd primes $\ell \le B$; twelve cutoffs as in the report. Output `data/recount.json` (sha256 `e84f89d3…7d98`; source sha256 `cb1dc02c…7550`). No $\ell \le 200000$ divides any n.

curve	split (obs.)	expected	sd	z @ 691	z @ 5,000	z @ 50,000	L-ratio, ℓ ≤ 200k
prime192v1 (P-192)	60	59.7	5.51	+0.05	+0.28	+0.80	0.93
prime192v2	58	59.7	5.51	−0.31	−2.20	+0.66	1.20

prime192v3	57	59.7	5.51	−0.49	−1.12	−1.51	0.80
secp224r1 (P-224)	50	60.1	5.53	−1.82	−2.54	−1.50	0.64
prime239v1	42	59.7	5.51	−3.21	−1.28	−0.12	0.56
prime239v2	57	59.7	5.51	−0.49	−0.89	−2.07	0.86
prime239v3	57	59.7	5.51	−0.49	+0.35	−0.51	0.93
prime256v1 (P-256)	57	60.2	5.54	−0.57	−0.30	−1.42	1.27
secp384r1 (P-384)	54	59.9	5.52	−1.06	−0.59	−1.21	0.62
secp521r1 (P-521)	52	60.3	5.54	−1.51	+0.54	+0.54	1.01

Counts at $B = 691$ (124 odd primes). Pooled z over the ten curves, $(\sum \text{obs} - \sum \text{exp})/\sqrt{\sum \text{var}}$, at B = 100, 200, 500, 691, 1000, 2000, 5000, 10000, 20000, 50000, 100000, 200000: −2.71, −1.81, −2.98, −3.13, −1.98, −2.16, −2.45, −2.12, −2.26, −2.00, −1.87, −1.79 (Stouffer $\sum z/\sqrt{10}$ agrees to two decimals). The last column is this study's normalisation of the partial Euler product $\prod (1 - \chi(\ell)/\ell)^{-1}$ over odd $\ell \le 200000$, as a ratio to $\exp E[\log \prod]$ under the conditioned model; it is not the report's "× typical" column, but the ordering agrees (prime239v1, P-384, P-224 lowest; P-256 highest).

Agreement with the uploaded report. Every split count at 691, every per-curve z at 691, 5,000 and 50,000, and every pooled z at the twelve cutoffs matches the report's table and list to the printed precision [pcc:unverified]. The report's random-curve calibration (4,267 SEA-counted curves) and its percentiles were not reproduced; no statement is made about them. Nothing here bears on the cause of the deficit; the report's own caveats (cutoff chosen after the fact, correlated cutoffs, min-of-three families, ten curves) stand.

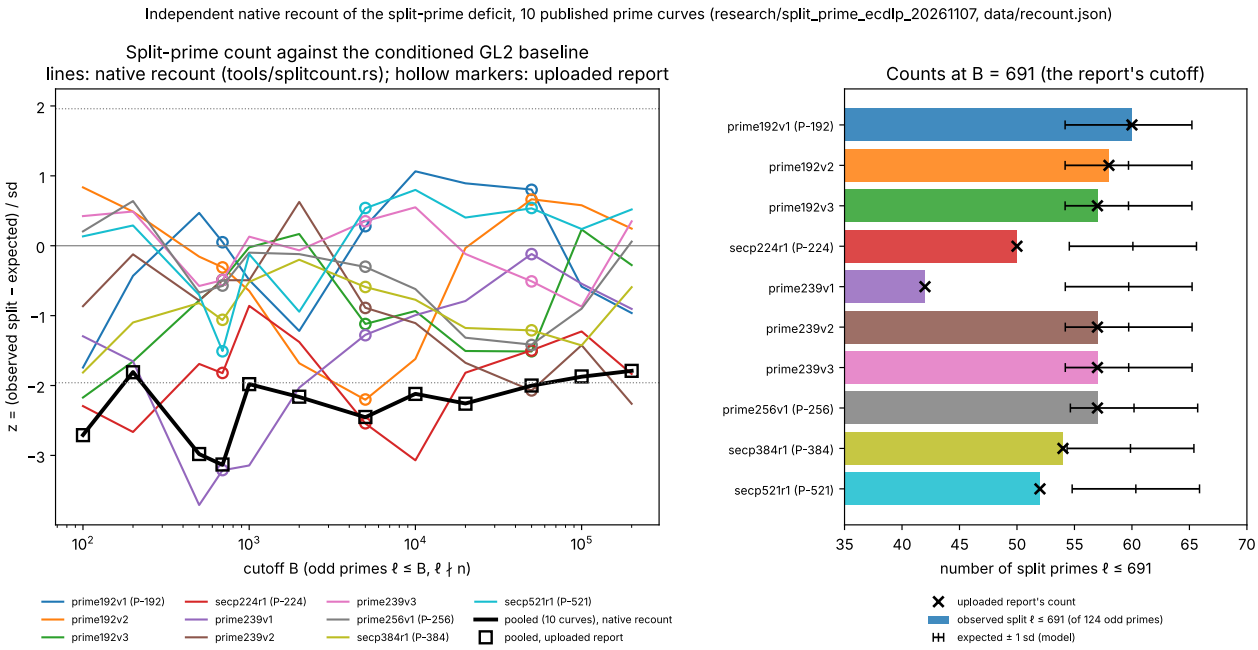


Figure 2: Left: z against cutoff for each curve (lines, native recount) with the report's values as hollow markers; black: pooled. Right: counts at $B = 691$ against the model expectation ± 1 sd. Source `figures/plot.py`, data `data/recount.json`; n = 124 primes at $B = 691$.

4. The one ECDLP-adjacent effect: horizontal expander degree

The JMV self-reduction walks the horizontal class along ℓ-isogenies for split $\ell \le B$; the graph has degree $2 \times \#\{\text{split } \ell \le B\}$. A deficit lowers the degree for a given B: at $B = 691$, P-224 has 50 split primes (degree 100) against 60 expected (degree 120), prime239v1 has 42 (degree 84). The walk needs a slightly larger B for the same mixing, i.e. a slightly larger transport cost; by §2 the destination curve is exactly as hard. The lab's protocol I-2 (`research/isogeny_class_difficulty_20261008/PROTOCOL-I2-transport-cost.md`) already prices transport; this study adds only the observation that the report's statistic is the degree of that graph.

5. Candidate novel directions that survive the obstruction

6. Status of every statement

statement	status
Split counts, expectations, z at twelve cutoffs for ten curves (§3)	measured natively from public parameters; agrees with the uploaded report; not independently checked by a second producer [pcc:unverified]
Theorem and corollary (§2)	derived (proof sketch); no Lean proof; no identity certificate (not a bilinear or summation identity)
"Model choice within the class is null"	measured by the lab on 2026-11-02 at 24–32 bits; cited, not re-measured
Expander-degree effect (§4)	derived from the counts; transport cost not measured here
Candidate directions (§5)	proposed; not run

Files. `tools/splitcount.rs`, `data/recount.json`, `data/uploaded_report_values.json`, `figures/plot.py` with `recount_z.{svg,png}` and `obstruction_diagram.{svg,png}`, `report/REPORT.typ` (this document), `problems/split-prime-pattern-scalar-only.md`. **Next steps.** A Lean statement of §2 (1)–(3); of the §5 candidates, the one with the best prior under a frozen protocol on 24–32-bit toy classes with the controls of `controlled-experiment`; the report's own pre-registered number-theory follow-ups only if the owner wants the fingerprint question answered.